Arvind Yogesh Suresh Babu

2485003276 | savyo@umich.edu | LinkedIn | GitHub | Portfolio

EDUCATION

University of Michigan **Aug 2025 - Dec 2026**
Master of Science in Data Science (GPA: 3.87/4) *Ann Arbor, MI*
- **Coursework:** Applied Machine Learning, Natural Language Processing, Statistical Inference, Advanced Database Management Systems, Advanced Probability & Distribution, Regression Analysis, Reinforcement Learning

Madras Institute of Technology **Aug 2020 - Jun 2024**
Bachelor of Engineering in Mechanical Engineering (GPA: 9.07/10) *Chennai, India*
- **Coursework:** Data Structures & Algorithms, Deep Learning, Software Engineering, Robotics Simulations

SKILLS

- **Languages**: Python, SQL, C++, Java, Go
- **ML/AI Frameworks**: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, MLflow, TensorBoard, PySpark, FAISS, Reinforcement Learning, Computer Vision, Adversarial ML, Langchain, RAG pipelines
- **MLOps / Infrastructure**: Docker, Terraform, Kafka, Spark, ONNX Runtime, AWS(S3, CloudFront, Elastic Beanstalk), Airflow, CI/CD(Github Actions), Git, Linux
- **Backend / Data**: FastAPI, Open3D, ROS , Vector Search, Tableau, MongoDB, Flask

WORK EXPERIENCE

Prof. Z. Morley Mao's Research Group, University of Michigan **Jun 2026 - Present**
Research Intern - AI Security & Machine Learning *Ann Arbor, United States*
- Built a modular adversarial machine learning attack framework using PyTorch, targeting vision-language-action (VLA) models for autonomous driving across two model backends (ReCogDrive, Alpamayo).
- Implemented differentiable Stanley and PID controllers and a kinematic bicycle model in Python to design closed-loop attack loss functions including steering command, curvature, and heading discontinuity.
- Designed a scene vulnerability analysis tool ranking autonomous driving scenarios by decision-boundary proximity using mode entropy and cluster separation metrics, validated with a 44-test automated unit test suite.

M-City, University of Michigan **Sep 2025 - May 2026**
Student ML Researcher *Ann Arbor, United States*
- Built and maintained automated ML evaluation pipelines (ROS, Open3D, Python) for batched model benchmarking, reducing manual testing effort by 63 percent.
- Supported dataset preparation, annotation quality checks, and fine-tuning for Co-DETR object-detection models; trained and evaluated CV models in PyTorch with MLflow/TensorBoard tracking for model comparison and distribution-shift analysis.

Vestas Wind Technology **Jun 2024 - Jul 2025**
Data Scientist Intern *Chennai, India*
- Developed time-series forecasting models on SCADA telemetry and engineered features, achieving 0.82 AUC with a median 18-day alert lead time for predictive maintenance.
- Built scalable ML workflows using Kafka, Spark, ONNX Runtime, and Airflow for anomaly detection, batch inference, and automated retraining, improving supply-planning accuracy by 56 percent - earned the Q1 Spot Award.

Computer Society, MIT **Feb 2021 - Jun 2024**
ML Engineer / Technical Lead
RoadSense
- Built a real-time streaming perception pipeline using Apache Kafka, Spark Structured Streaming, and CUDA-accelerated PyTorch inference, containerized across four microservices with Docker Compose and Prometheus monitoring.
- Benchmarked Faster R-CNN and SSDLite MobileNetV3 on an NVIDIA H200 GPU, achieving 9.9ms and 14.9ms median latency (99.6 and 64.2 FPS), both clearing the 150ms real-time target.

PROJECTS

Synapse — Open-Source LLM Gateway **May 2026 - Aug 2026**
- Built and open-sourced an **OpenAI-API-compatible** LLM gateway (FastAPI, React, Redis, Postgres) adding semantic response caching (**Redis ANN vector search**), streaming, per-key rate limiting, and cost/latency observability — verified drop-in compatibility with the official SDK via automated tests, not just schema comparison.
- Designed a **self-tuning cache-correctness controller** using LLM-judge shadow verification to auto-calibrate similarity thresholds, holding precision at **98–100%** under real GPU-served inference (Llama 3.1 8B, H200) while identifying and fixing two measurement bugs (EWMA smoothing, benchmark ground-truth labeling) surfaced during development.

Attribution-Guided Masking for Robust Cross-Domain Sentiment Classification **Jan 2026 - Apr 2026**
- Co-developed an attribution-guided regularizer that reduced cross-domain generalization gap to 0.013–0.244 across 4 sentiment domains (IMDb, Amazon, Hotel, Sentiment140), beating DANN, IRM, Group DRO, and Fish on 3 of 4 zero-shot transfer targets.
- Ran a strict zero-shot evaluation with bootstrap 95% CIs across 12 domain-transfer pairs; ablation isolated attribution-guided masking as the driver, reducing generalization gap by 0.017 absolute over random masking.